

Erdős Problem #413

EP-413

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\omega(n)$ count the number of distinct primes dividing n . Are there infinitely many n such that, for all $m < n$, we have $m + \omega(m) \leq n$?

Can one show that there exists an $\epsilon > 0$ such that there are infinitely many n where $m + \epsilon\omega(m) \leq n$ for all $m < n$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>